

CUSTOMER CHURN

Analysis & Retention Strategy (IBM Telco Dataset)

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Project Overview



Business Problem

Telecommunication companies lose customers regularly, which affects revenue and long-term growth.



Objective

Identify the major customer behaviors that predict churn and recommend strategies to improve retention.



Key Focus Areas

- Customer tenure
- Contract type
- Monthly charges
- Service usage

Dataset Overview

DATA SOURCE

IBM Telco Customer Churn Dataset
(Kaggle)

SIZE

7,000+

RECORDS

21

COLUMNS

DATA INCLUDES

-  Customer demographics
-  Services subscribed to
-  Billing/payment information
-  Contract details
-  Churn status

Tools and Data Cleaning

TOOLS USED



Excel



MySQL



Power BI



R Online

DATA CLEANING STEPS

- ✓ Removed duplicates
- ✓ Checked missing values
- ✓ Converted TotalCharges to numeric
- ✓ Standardized categorical fields
- ✓ Removed blank spaces and formatting issues

Key Analysis Areas

— QUESTIONS INVESTIGATED

01

Which contract types have the highest churn?

02

Does customer tenure affect churn?

03

Do higher monthly charges increase churn?

04

Which services reduce churn risk?

05

Which payment methods are associated with churn?



Key Findings

— MAJOR INSIGHTS



Month-to-month contracts had the highest churn rate



Customers with shorter tenure churned more often



Higher monthly charges increased churn likelihood



Customers without tech support and online security churned more

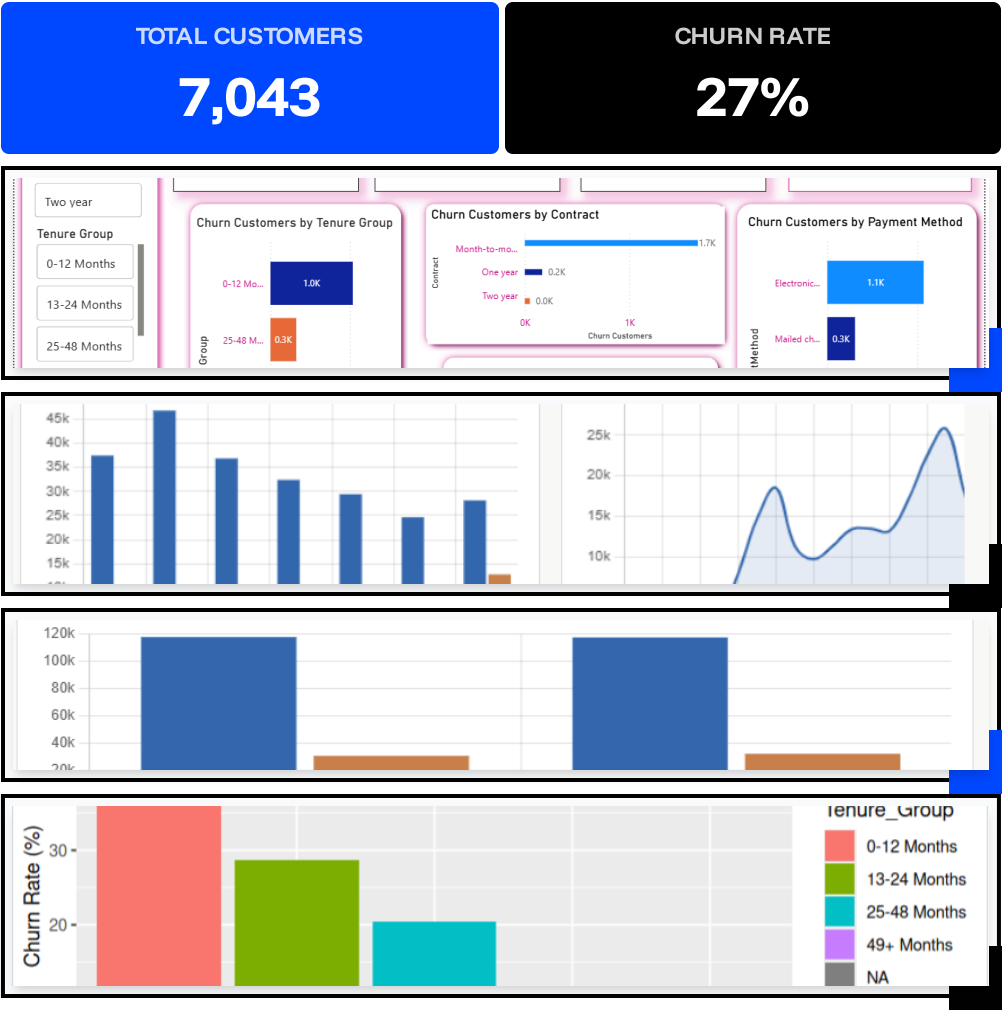


Electronic check users showed higher churn rates

82%

INSIGHT CORRELATION

Dashboard Visualizations



COMPONENTS



KPI Cards



Churn Rate Gauge



Contract Type Bar Chart



Tenure Analysis



Monthly Charges Comparison

Recommendations

1

Encourage long-term contracts

2

Improve onboarding for new customers

3

Bundle support-related services

4

Reward high-value customers

5

Promote automatic payment methods

Conclusion

FINAL SUMMARY

Contract type, tenure, monthly charges, and support services strongly influence customer retention.